Forecasting One-to-Three-Day Discharge and Flood Risk of the Huong River at Kim Long as a Local Correction Layer on GloFAS

Dang Nguyen Giap Ho Anh Duc Mai Thi Le Huyen

Department of Information Technology, FPT University
DSP391m — Data Science Project — Final Report, Fall 2026
Hue, Vietnam

***Abstract*—**We forecast the daily GloFAS v4 discharge of the Huong River at Kim Long, Vietnam, one to three days ahead from open data only (107.5 million hourly records and daily records since 1997) and translate it into flood-risk levels defined as flood-season percentiles, since discharge does not determine the official water-level alert stages. A LightGBM model with 88 predictors, including storm, cold-surge and ENSO indicators, reaches NSE 0.747, 0.499 and 0.236 on a test period from July 2022 (persistence 0.542, 0.008, –0.198) and 0.761, 0.437 and 0.206 in a 13-fold yearly walk-forward; the three-day target of 0.25 is missed. One day ahead it catches 28 of 43 flood events, as many as persistence but with lead time, and isotonic calibration makes its alert probability reliable (Brier 0.0119 to 0.0100) so that the Bayes threshold 1/11 can be used directly. Peaks are cut; peak-weighted training, isotonic correction and an LSTM ensemble did not meet pre-set criteria. Reservoir effects are absent from the target, satellite rainfall helps only in hindsight, deep learning does not beat LightGBM across years, and a Spark cluster reproduces the pipeline bit for bit but is 11.6 times slower than Polars at this scale. The model runs daily on GitHub Actions and feeds a Streamlit dashboard.

***Keywords*—**flood forecasting, GloFAS post-processing, LightGBM, walk-forward validation, probability calibration, Apache Spark, LSTM, Huong River

I. INTRODUCTION

Central Vietnam is the part of the country most exposed to river floods. Hue lies on the Huong River, whose catchment to the Kim Long station is short and steep: rain from tropical cyclones, cold surges and the north-east monsoon becomes a flood peak within a day [5]. Official warnings are issued as water-level alert stages (1.0, 2.0 and 3.5 m at Kim Long [4]), while the only open, long, daily discharge record for the basin is a simulation: the Global Flood Awareness System (GloFAS) v4 [1], [2], driven by ERA5 [3].

This project forecasts the daily GloFAS discharge at the Kim Long cell (16.45° N, 107.50° E) one, two and three days ahead from open data only, and translates the forecast into flood-risk levels. It is positioned as a *local correction layer* on top of GloFAS, in the line of recent GloFAS post-processing studies [7], [8], not as a replacement: the global model is known to degrade in small, steep, monsoon-driven catchments, and a local model can add predictors that the global system does not use (storms, cold surges, soil moisture, local rainfall, reservoir orders).

The work was delivered in four stages: a proposal (Report 1), data collection and exploratory analysis (Report 2), modelling and evaluation (Report 3), and distributed processing with deep learning (Report 4). This final report merges them and adds the work completed after Report 4. Its contributions are:

- a reproducible open-data pipeline (GloFAS, ERA5, ERA5-Land, IBTrACS, ONI, IMERG, GSMaP and GSMaP_NRT; about 107.5 million hourly cell records) with automated leakage tests;
- a LightGBM model with 88 predictors (configuration E) that beats persistence, climatology, seasonal naive and ARIMA at every horizon, evaluated on a held-out period and in a 13-fold yearly walk-forward;
- risk levels defined as flood-season percentiles of discharge, a cost-based alert threshold and isotonic calibration of the alert probability;
- negative results reported as such: reservoir effects absent from the target, satellite rain helpful only in hindsight, deep learning and peak-weighting not better than LightGBM, Spark slower than a single machine at this data size;
- an operational daily job on GitHub Actions and a Streamlit dashboard.

Report 1 posed three research questions: (RQ1) can daily discharge be predicted one to three days ahead from gridded rainfall and antecedent discharge, and with what error; (RQ2) with what detection rate and false-alarm ratio can a classifier flag days above an official alert stage; and (RQ3) which sub-basins carry the highest flood risk. Table I gives the analytics position, unchanged since the proposal. Section XI answers the three questions.

TABLE I
POSITION ON THE FOUR CLASSES OF ANALYTICS

Class	Position	Reason
Descriptive	Supporting	Seasonality, lag structure and flood events establish that the signal exists (Report 2).
Diagnostic	Partial	SHAP attribution and the reservoir tests explain forecasts; no causal design is claimed.
Predictive	**Core**	Discharge and risk level one to three days ahead.
Prescriptive	Out of scope	Evacuation advice needs authority and accountability that an academic project does not hold.

II. RELATED WORK

Global flood forecasting. GloFAS couples a land-surface and routing model to ECMWF forcing and provides daily discharge worldwide on a ~5 km grid [1], [2]. Skill is lower in small, steep catchments, which motivates local post-processing; ensemble machine-learning bias correction of GloFAS has recently been shown to improve station-level flow [7], [8].

Machine learning for streamflow. Gradient-boosted trees such as LightGBM [10] are strong on tabular hydrological predictors and train in seconds at this data size. LSTM networks learn rainfall–runoff behaviour from sequences [11], [12] and are the state of the art for global ungauged flood prediction [6], but their advantage is clearest when many catchments are pooled; transfer to a single small basin is an open question [13].

Evaluation and explanation. The Nash–Sutcliffe efficiency (NSE) [14] and the Kling–Gupta efficiency [15] are the reference skill scores; guidance places satisfactory streamflow simulation near NSE 0.5 [16]. Rare events are scored with contingency measures (POD, FAR, CSI) and the Brier score [17]; isotonic regression is a standard way to calibrate classifier probabilities [18]. SHAP values [19] attribute tree-ensemble forecasts to their inputs and are used for streamflow models [9].

Distributed processing. Apache Spark [20] and MLlib [21] distribute data and model fitting; PyTorch DDP [22] distributes neural-network training. The COST argument [23] warns that a distributed system should be compared with a competent single-machine implementation before claiming a speed-up.

III. STUDY AREA AND DATA

A. Basin

The Huong catchment to Kim Long covers about 1,540 km² (1,543 km² delineated from the Copernicus GLO-30 DEM [24]). Report 1 used 2,830 km², which is the whole Huong system including the Bo River (938 km²) that joins downstream of Kim Long; Report 3 corrected this. A D8 flow model splits the catchment into five sub-basins (Fig. 1): the Ta Trach and Binh Dien reservoir catchments (670 and 507 km²; capacities 646 and 424 million m³ [25]), the reaches below the dams (88 and 185 km²) and the reach from the Tuan confluence to Kim Long (93 km²).

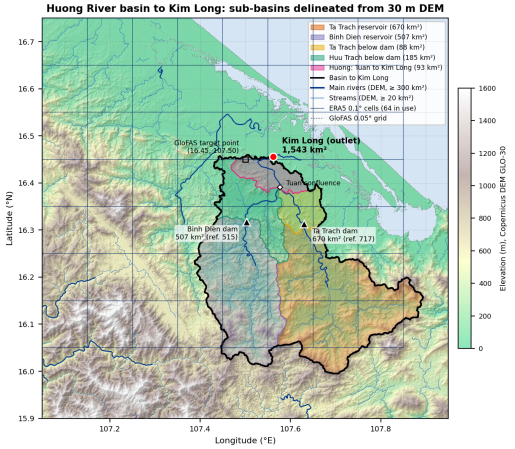


Fig. 1. Huong basin to Kim Long (1,543 km²), five sub-basins from the 30 m DEM, ERA5 cells and the GloFAS target point.

B. Target: GloFAS v4 discharge

The target is the GloFAS v4 daily mean discharge at the cell selected in Report 1 by three independent tests (distance to OpenStreetMap river lines, correlation between candidate cells, and the catchment implied by specific runoff). Through Open-Meteo [26] the series starts in 1997 for this basin. Three caveats bound every result:

- **It is a simulation.** All scores measure agreement with GloFAS, not with the river.
- **Regime change in July 2022.** Before 1 July 2022 the series is reanalysis; from that date it is built from archived forecasts. Report 2 found no systematic shift in the heavy-rain response (Mann–Whitney p = 0.581), but only 3 full walk-forward years fall in the new regime, too few to test it alone.
- **No dams and under-simulated floods.** Four flow indices show no change after the Ta Trach and Binh Dien dams came into operation (all p above 0.05; Section V-H), and reported natural inflows to the two reservoirs exceed the simulated discharge at Kim Long on 14 of 23 bulletin days (Fig. 7).

C. Predictors

Table II lists the inputs. ERA5 rainfall on 64 cells (6,544,896 hourly rows since 2015, daily since 1997) is averaged per basin band and sub-basin. ERA5-Land adds soil moisture in three layers; IBTrACS v04r01 [27] adds 417 tropical cyclones within 1,500 km of Hue; ERA5 pressure and wind give cold-surge indicators; the Oceanic Niño Index [28] is lagged three months so that only published values are used; and 15-day, 30-day and season-cumulative rainfall since 1 September act as proxies for reservoir storage. Satellite rainfall (IMERG V07 [29] and GSMaP v8 [30] from Earth Engine, and GSMaP_NRT from JAXA) was retrieved hourly on 100 cells since 1998. With ERA5, the hourly inputs total about 107.5 million cell records (101.0 million from the three satellite products).

TABLE II
OPEN DATA USED BY THE PROJECT

Source	Variables	Period, step	Role
GloFAS v4 [1], [26]	Daily discharge, one cell	1997–2026, 1 day	Target, lags
ERA5 [3]	Rain, met. (64 cells, 0.1°)	1997–2026, 1 h from 2015	Main predictors
ERA5-Land	Soil moisture, 3 layers	1997–2026, 1 day	Config. C
IBTrACS [27]	Cyclone distance, wind, speed	1996–2026	Config. E
ERA5 pressure, wind	Cold-surge indices	1997–2026	Config. E
ONI [28]	ENSO, lagged 3 months	Monthly	Config. E
IMERG, GSMaP [29], [30]	Hourly rain, 100 cells	1998–Oct. 2026	Tested
GSMaP_NRT (JAXA)	Hourly rain, low latency	1998–Oct. 2026	Tested
Bulletins, news	132 reservoir records	2016–2026	Checks, floor rule

D. Exploratory findings (Report 2)

The 2010–2026 daily panel has 6,087 days with no missing discharge or rainfall. Discharge is highly skewed (median 49.7 m³/s, maximum 3,588 m³/s on 8 November 2011, skewness 5.91), so models are trained on log(1 + Q). September to December carry 58.6% of the runoff. Rainfall correlates best with discharge one day later (r = 0.789), which supports a short forecast horizon, and wet antecedent conditions matter: after heavy rain the median discharge is 585.1 m³/s on wet ground against 84.9 m³/s on dry ground.

E. Data engineering

Acquisition is a set of independent tasks, each writing one Parquet file; a task whose output exists is skipped, so an interrupted crawl resumes without re-requesting data, request pacing adapts to rate-limit replies, and a lock file prevents concurrent crawlers (three parallel crawlers had once been mistaken for provider throttling). Raw files are converted to Vietnam local time (UTC+7), hourly rain is aggregated to days with a test that the hourly values sum to the daily value, and validation functions raise errors instead of silently repairing negative discharge, duplicated dates or gaps. Satellite rain was cut to a box of cells around the basin as it was downloaded (Earth Engine month by month, GSMaP_NRT from JAXA's server). Raw data are backed up as a checksummed GitHub release, and every experiment writes versioned CSV files from which the numbers of this report are recomputed and checked when the PDF is built.

The rainfall sources disagree. Over the 2015–2025 flood seasons IMERG, GSMaP and gauge-calibrated GSMaP total 0.90, 0.74 and 1.14 times ERA5. ERA5 correlates best with next-day GloFAS discharge (r = 0.78 against 0.69 for IMERG), as expected because GloFAS is driven by ERA5; this does not show that ERA5 is closer to the true rain. PERSIANN-CCS, the only product finer than GloFAS (0.04°), recorded 0.073 of the ERA5 total in November 2020, probably missing rain from low, warm monsoon clouds, and was not used.

F. Why no gauge data

Observed hourly water levels and discharge at Kim Long belong to the national hydro-meteorological service. The team decided not to request them, so the target stays the open GloFAS daily series and hourly or finer data enter only as inputs. The official alert stages are water levels, and the attempt to map discharge to level from published flood peaks failed: across 12 floods log discharge explains little of the peak level (R^2 = 0.18), and the best fit missed the five floods after July 2022 by –1.94 m on average. Tide and backwater from the Tam Giang lagoon, the Thao Long barrage and reservoir operation decouple level from simulated discharge.

IV. METHODOLOGY

A. Task, features and leakage control

Each forecast is issued at the end of day t for h = 1, 2 and 3 days. Every predictor is known at the end of day t: rolling windows look backward only, and automated tests fail if a feature uses later information or if NSE exceeds 0.98, which would signal leakage. Five configurations add data cumulatively: A is the Report 2 panel (66 features: discharge lags of 1–14 days, band rainfall with lags and sums, an antecedent precipitation index, season code) trained from 2010; B trains from 1997; C adds soil moisture (70); D adds reservoir-storage proxies (73); E adds storm, cold-surge and ENSO predictors (88).

B. Splits and walk-forward protocol

Single regime split: train to December 2015, validation to June 2022, test from 1 July 2022 (1,522 days at h = 1); final models are refit on train plus validation. **Yearly walk-forward:** each year 2013–2025 is a test fold (13 folds) with an expanding training window, a 3-day gap so that no h = 3 label enters the test year, and risk thresholds recomputed from each fold's training data. Gains are reported as the mean ΔNSE over folds with a 95% bootstrap interval over folds, folds won/lost and a two-sided Wilcoxon test. Perturbing the inputs at the 13th decimal moves NSE by about ±0.01, so smaller differences are not resolvable.

C. Main model: LightGBM configuration E

LightGBM [10] is trained on log(1 + Q) with 600 trees, learning rate 0.05, 31 leaves and at least 30 days per leaf, one model per horizon, averaged over seeds 42, 7 and 2026. The loss is L1 at h = 1 and L2 at h = 2–3. The switch was chosen on the walk-forward and the validation period, never on the test set: L2 raised the mean NSE by +0.039 [+0.020; +0.057] at h = 2 and +0.036 [+0.020; +0.051] at h = 3 (folds won/lost 11/2, p = 0.003 and 0.002), while at h = 1 the gain (+0.018, p = 0.22) was not significant and validation fell (–0.011). The model was frozen on 02/11/2026 after the experiments of Section VII.

D. Baselines

Persistence, climatology of the calendar day (training years only), seasonal naive and ARIMA(2,1,2) on log discharge. The raw GloFAS forecast cannot be rebuilt for past dates: Open-Meteo does not archive past flood forecasts (HTTP 404), and the GloFAS analysis at t + h would leak the target. Issued GloFAS forecasts have therefore been logged every morning since 15 September 2026.

E. Risk levels instead of alert stages

Because discharge does not determine the official water level (Section III-E), labels are *risk levels* at the 95th, 98th and 99.5th percentiles of September–December discharge on train plus validation: 806, 1,275 and 1,905 m³/s. They are relative to the GloFAS climate and are *not* the official alert stages. The test period holds 31 level-1 days, 12 level-2 and 7 level-3 days; a level supports a main conclusion only with at least 30 positive days, so level 1 is the operational level.

F. Classifier, cost-based threshold and calibration

A binary LightGBM classifier per level and horizon is weighted by the negative-to-positive ratio instead of resampling, which would break the temporal structure. A fixed threshold of 0.5 ignores that a missed flood costs more than a false alarm, so the threshold minimises C = 10 × misses + false alarms (sensitivity 5:1 and 20:1). It is chosen on out-of-fold probabilities of earlier years only. Isotonic regression [18] is fitted on the same pool (years before the test year) and maps raw scores to calibrated probabilities; on a calibrated scale the Bayes threshold for a 10:1 cost ratio is 1/(1 + 10) = 0.091.

G. Reservoir floor rule

When the province orders releases, the forecast at Kim Long becomes the larger of the model forecast and the sum of ordered releases at Ta Trach and Binh Dien, since released water reaches Kim Long within hours. The rule only raises a forecast. It is a safeguard, not a validated component: no historical series of orders exists to score it.

V. RESULTS

A. Baselines and LightGBM

LightGBM beats every baseline at every horizon (Table III). Configuration E reaches NSE 0.747, 0.499 and 0.236 on the test period, against 0.542, 0.008 and –0.198 for persistence, and the yearly walk-forward gives a consistent picture (0.761, 0.437, 0.206; persistence 0.536, –0.084, –0.361). **Acceptance criterion AC-1** (NSE at least 0.70, 0.40 and 0.25) is met at h = 1 and h = 2 and **missed at h = 3**: 0.236 on the test period and 0.206 in the walk-forward, both below 0.25. The RMSE criterion at h = 1 is met (25.6% below persistence for configuration A, target 20%). Skill decays quickly beyond one day because the rain that drives a flood two or three days ahead has usually not yet fallen at issue time.

TABLE III
TEST PERIOD JULY 2022 – AUGUST 2026. CONFIG. A: L1, SEED 42; CONFIG. E: MEAN OF THREE SEEDS, RMSE NOT RECORDED (N/R)

Model	NSE h=1	h=2	h=3	RMSE h=1 (m³/s)	h=2	h=3
Persistence	0.542	0.008	–0.198	156.4	230.2	253.1
Climatology	0.097	0.097	0.096	219.6	219.7	219.8
Seasonal naive	–0.434	–0.434	–0.435	276.8	276.8	276.9
ARIMA(2,1,2)	–0.093	–0.093	–0.093	241.6	241.7	241.8
LightGBM, config. A	0.747	0.420	0.175	116.3	176.1	210.0
LightGBM, config. E (main)	**0.747**	**0.499**	**0.236**	**n/r**	**n/r**	**n/r**

B. Does more data help?

A longer record (B) is the only change that helps at h = 1 (+0.021; Table IV). Soil moisture (C) and storage proxies (D) give small changes of mixed sign: that information is already carried by discharge lags and the API. Storm, cold-surge and ENSO predictors (E) give the largest gain at h = 2 and 3 (+0.038 and +0.037 over D): these drivers announce heavy rain one to three days ahead. The seed standard deviation is at most 0.012.

TABLE IV
CUMULATIVE ABLATION, TEST PERIOD, MEAN OF THREE SEEDS

Config.	Added	NSE h=1	h=2	h=3
A	Report 2 panel, from 2010 (4,550 days)	0.745	0.432	0.166
B	Record from 1997 (9,298 days)	0.766	0.447	0.192
C	+ soil moisture	0.757	0.454	0.208
D	+ reservoir-storage proxies	0.756	0.461	0.199
E	+ storms, cold surges, ENSO	0.747	0.499	0.236

Two further tests bound what better rainfall can give. Nine hourly ERA5 features per basin (own split, as hourly ERA5 starts in 2015) raise NSE at h = 3 from 0.168 to 0.183, consistently across seeds (+0.016, +0.008, +0.022), while hourly features per sub-basin are no better than the basin mean (0.184): ERA5 is too coarse to separate them. Multiplicative log-normal noise with $\sigma = 1$ on all rainfall inputs lowers NSE at h = 1 from 0.742 to 0.683 and not at all at h = 3 (0.167 to 0.172). The model leans on discharge lags, so imperfect rainfall costs little; for the same reason it cannot anticipate rain that has not yet fallen.

C. Satellite rainfall and latency

On the single split IMERG appeared to help at h = 2; across 13 folds the only consistent gain on configuration E comes from gauge-calibrated GSMaP at h = 3 (+0.018 [+0.008; +0.028], positive in 11 of 13 folds, p = 0.008; Table V, Fig. 2). The Earth Engine archive, however, arrives about 27 h late, so a forecast issued near 05:00 has satellite rain only up to day t – 1; with features lagged one day the gain disappears (–0.001, p = 0.839). JAXA's GSMaP_NRT arrives about 4.7 h after

observation and was tested by retraining on its own history: it helps the weaker Report 2 panel at every horizon (+0.009, +0.019, +0.012) but adds nothing to configuration E (–0.002, +0.003, –0.001; all p at least 0.27). Satellite rain partly substitutes for predictors that E already contains; it was not added to the operational model.

TABLE V
ΔNSE FROM SATELLITE RAINFALL, 13 YEARLY FOLDS: MEAN, 95% INTERVAL, FOLDS WON/LOST, WILCOXON p

Config. (vs. own baseline)	h = 1	h = 2	h = 3
E + GSMaP-GC, same day	–0.001 (8/5)	+0.012 (8/5), p 0.38	+0.018 [+0.008; +0.028] 11/2, p 0.008
E + GSMaP-GC, lagged one day	+0.001 (8/5)	+0.004 (7/6)	–0.001 [–0.007; +0.005] 8/5, p 0.839
E + GSMaP_NRT	–0.002 (4/9)	+0.003 (6/7)	–0.001 [–0.009; +0.008] 5/8, p 0.54
A + GSMaP_NRT	+0.009 (10/3), p 0.008	+0.019 [+0.008; +0.031] 11/2, p 0.006	+0.012 [+0.005; +0.019] 12/1, p 0.005

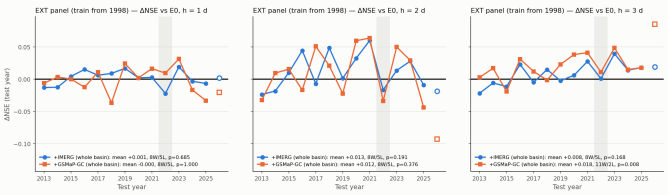


Fig. 2. ΔNSE per test year when IMERG or gauge-calibrated GSMaP is added to configuration E (three seeds). Shaded: GloFAS regime change (07/2022); hollow markers: partial year 2026, excluded.

D. Risk-level detection and flood events

The level-1 classifier detects 20 of the 31 level-1 test days one day ahead with 5 false alarms (POD 0.645, FAR 0.200, CSI 0.556; Table VI); CSI falls to 0.268 at h = 2 and 0.087 at h = 3. Level 2 has only 12 positive test days, so criterion AC-2 (POD at least 0.7 for the second level) can be neither met nor evaluated reliably.

TABLE VI
RISK-LEVEL CLASSIFIERS ON THE TEST PERIOD (THRESHOLD 0.5)

Level	h	Positive test days	POD	FAR	CSI	PR-AUC
1 (806 m³/s)	1	31	0.645	0.200	0.556	0.671
	2	31	0.355	0.476	0.268	0.354
	3	31	0.129	0.789	0.087	0.153
2 (1,275 m³/s)	1	12	0.333	0.500	0.250	0.419
	2	12	0.000	1.000	0.000	0.082
	3	12	0.000	1.000	0.000	0.035

Scored per flood *event* (consecutive level-1 days), the main regression model catches 28 of 43 walk-forward events one day ahead, level with persistence (27; Fig. 3), and has real lead time: 15 caught events were flagged before their onset, which persistence can never do. At h = 2 it catches 10 against 14 for persistence and at h = 3 none against 7, with almost no false events (1 and 0 against 16 and 29). The reason is peak cutting: the forecast peak is a median 0.80, 0.54 and 0.31 of the simulated peak. On the 13 test-period events the counts are 6, 3 and 0 against 9, 3 and 3 for persistence; the walk-forward refits every year, the single split does not.

E. Alert threshold and calibration

Across 13 folds (4,748 days, 104 level-1 days at h = 1) the cost-based threshold cuts misses from 32 to 13 at the price of 8.4 false alarms per year instead of 3.1, lowering the 10:1 cost from 360 to 239 (p = 0.042; Table VII). Raw probabilities are badly scaled: the optimal raw threshold is only 0.03. Isotonic calibration does not change the alert cost (230; difference –0.7 per fold, p = 0.75), because a monotone map keeps the ranking of days. What it improves is the probability itself: the Brier score falls from 0.0119 to 0.0100 (skill score 0.44 to 0.54, ΔBrier –0.0020 [–0.0036; –0.0006] per fold, p = 0.027, better in 9 folds) and the expected calibration error from 0.010 to 0.002 (Fig. 4). After calibration the selected threshold (0.12, range 0.10–0.15) is close to the Bayes value 0.091, and using 0.091 directly gives the same cost (230) without any selection. The operational job therefore displays calibrated probabilities and alerts at 1/11. At 5:1 calibration helps slightly (168 to 157), at 20:1 it costs more (352 to 386).

TABLE VII
LEVEL-1 ALERTS AT h = 1, CONFIGURATION E, 13 WALK-FORWARD FOLDS (THRESHOLD: MEDIAN AND RANGE OVER FOLDS)

Rule	Threshold	Misses	False alarms (per year)	POD / FAR / CSI	Cost 10:1
Fixed 0.5, raw	0.50	32	40 (3.1)	0.69 / 0.36 / 0.50	360
Cost-based, raw	0.03 (0.02–0.04)	13	109 (8.4)	0.88 / 0.55 / 0.43	239
Cost-based, isotonic	0.12 (0.10–0.15)	12	110 (8.5)	0.88 / 0.54 / 0.43	230
Bayes 1/11, isotonic	0.091	12	110 (8.5)	0.88 / 0.54 / 0.43	230

F. Peak underprediction

Peaks are cut at every horizon: the mean peak bias of configuration E on the test period is –0.46, –0.71 and –0.86. For the forecast issued on 15 October 2022 the model gave 862 m³/s against a GloFAS peak of 1,967 m³/s; on 30 October 2025 it gave 1,727 against 2,903 m³/s. The limit is structural: only 12 of the 9,283 training days exceed the 2022 peak, each leaf holds at least 30 days, the L1 loss targets a conditional median, and every training day above 2,000 m³/s receded the next day, so the model predicts recession while the river is still rising. Section VII tests three remedies.

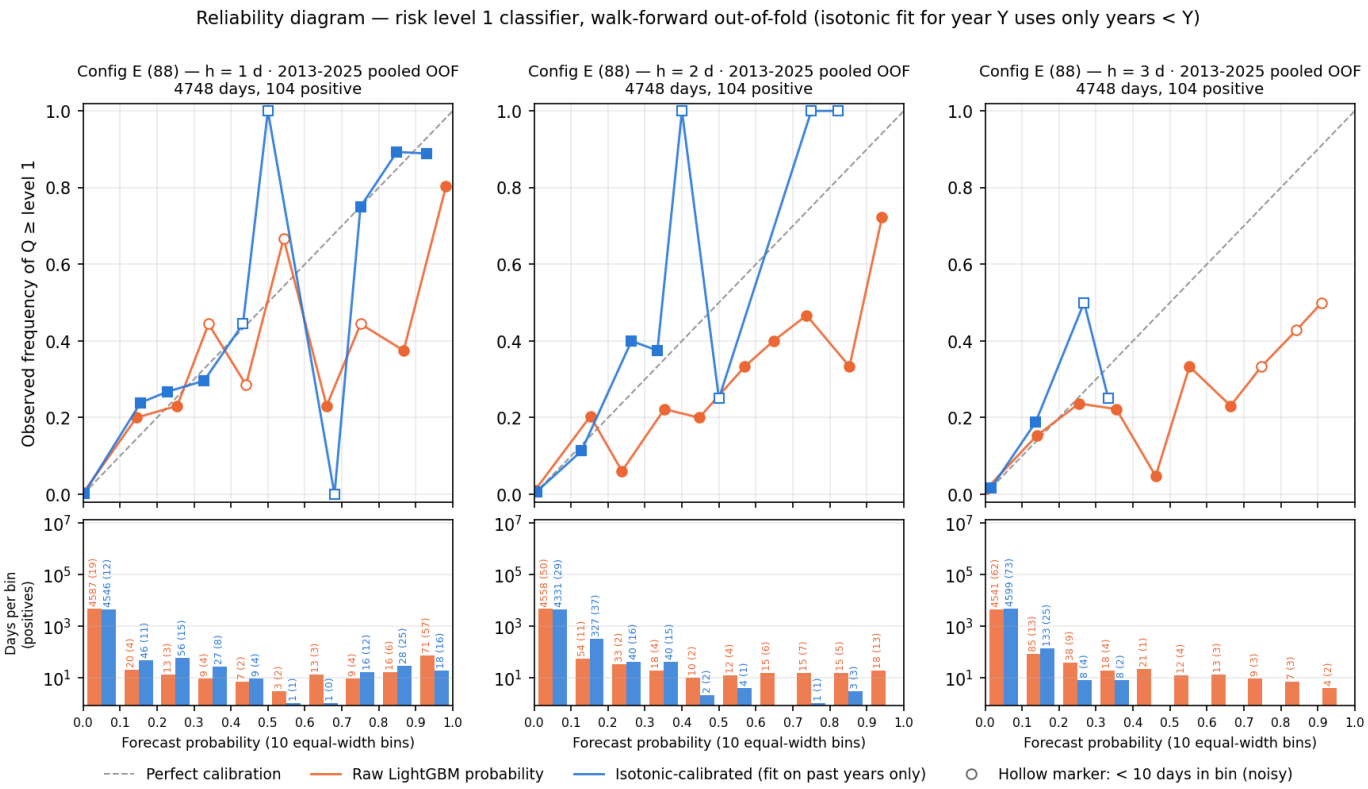


Fig. 4. Reliability of the level-1 classifier, configuration E, pooled out-of-fold probabilities 2013–2025 (h = 1, 2, 3). Orange: raw LightGBM; blue: isotonic calibration fitted on earlier years only; bars: days per bin (log scale); hollow markers: fewer than ten days.

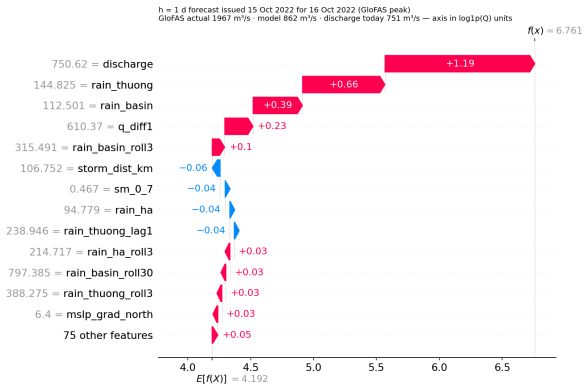


Fig. 5. SHAP waterfall (log(1 + Q) units) for the forecast issued on 15 October 2022: discharge and rainfall push the forecast up, but it stays far below the simulated peak of the next day.

G. Explainability

Tree SHAP on configuration E (Fig. 6) shows a plausible core. Today's discharge dominates at h = 1 (37.4% of mean |SHAP|); with lead time the weight moves from lagged discharge (58.4% to 22.8% by group) to rainfall (largest at h = 2, 36.5%) and to large-scale drivers: the pressure, wind and humidity group grows from 2.3% to 16.3%, and the north–south pressure gradient that marks cold surges ranks third at h = 3. Two signals are not plausible: soil moisture acts in reverse (Spearman correlation with its SHAP value –0.25 at h = 2), because it duplicates wetness already in today's discharge, and the full-reservoir proxy is not supported. At h = 3 the calendar carries 15.6%: when the dynamical signal is weak the model falls back on the season.

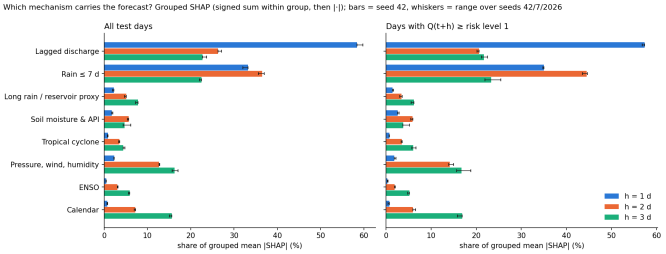


Fig. 6. Share of grouped mean |SHAP| by mechanism and horizon, configuration E, all test days (left) and level-1 days (right); whiskers: range over three seeds.

H. Reservoirs

If GloFAS represented the Ta Trach and Binh Dien reservoirs, the 30-day low flow, runoff ratio, flashiness and storm response after both dams operated (2014–2026) should differ from 1998–2008. None changes significantly (low flow 29.7 to 26.5 m³/s, p = 0.42; flashiness 0.435 to 0.441, p = 0.64). Bulletins show the opposite problem: on 15 October 2022 the dams released at least 3,672 m³/s with an inflow of 13,787 m³/s, while GloFAS gave 1,967 m³/s. The target is an unregulated river that under-simulates the largest floods; reservoir effects cannot be learned from it, which motivates relative risk levels and the floor rule.

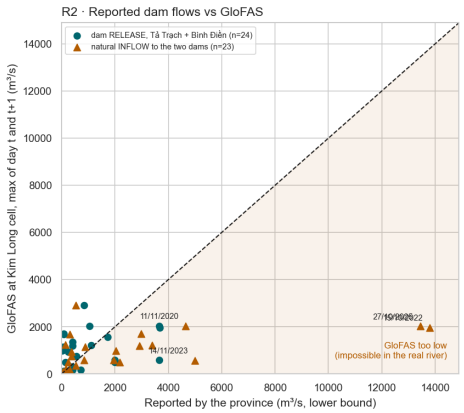


Fig. 7. Reported dam releases and natural inflows (lower bounds) against GloFAS at Kim Long. Points below the diagonal are impossible for the real river.

I. Acceptance criteria

Table VIII sets the results against the criteria fixed in the specification before modelling. AC-5 is met; AC-3 and AC-6 are met in part (against persistence, and for level 1 only); AC-1 is met at two of three horizons; AC-2 could not be evaluated and AC-4 failed. The reasons are stated rather than the criteria being relaxed.

TABLE VIII
ACCEPTANCE CRITERIA OF THE SPECIFICATION AND FINAL STATUS (TEST PERIOD UNLESS STATED)

Criterion	Target	Result	Status
AC-1 regression	NSE 0.70 / 0.40 / 0.25; RMSE 20% below persistence at h = 1	E: 0.747 / 0.499 / 0.236 (walk-forward 0.761 / 0.437 / 0.206); RMSE 25.6% below (A)	Met at h = 1, 2; **missed at h = 3**
AC-2 detection	POD at least 0.7 for the second level, h = 1	POD 0.333 on 12 test days	Not evaluable
AC-3 baselines	RMSE not above the best baseline at every horizon; raw GloFAS comparison	RMSE 116.3 / 176.1 / 210.0 against 156.4 / 230.2 / 253.1	Met; GloFAS part pending (log)
AC-4 level mapping	Historic floods within one alert stage	R^2 = 0.18; bias –1.94 m after 07/2022	Failed; replaced by risk levels
AC-5 no leakage	Leakage tests pass; NSE below 0.98	All tests pass; highest NSE 0.747	Met
AC-6 sample size	At least 30 positive test days for a main conclusion	Level 1: 31; level 2: 12	Met for level 1 only

VI. DISTRIBUTED PROCESSING AND DEEP LEARNING

A. Spark cluster and pipeline parity

A Spark standalone cluster (one master, 2 workers with 4 cores and 2 GB each, Spark 4.1.3, Java 21, CPU PyTorch 2.14.1) runs in Docker on one machine; the repository mounted at `/opt/repo` plays the role of HDFS or S3. The ETL turns 6,544,896 hourly ERA5 rows into 272,704 cell-days with the same result as the Polars pipeline, and the 54 hourly rainfall features agree with Polars bit for bit (613,584 hourly and 230,064 daily cells). Exactness required summing in the Polars order and taking the sine of the peak hour from a Polars table, because one unit in the last place can flip a threshold test.

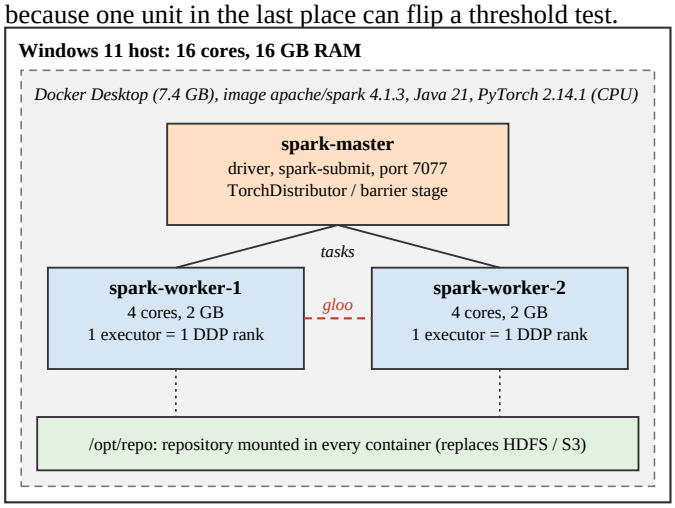


Fig. 8. Spark standalone cluster used for the distributed experiments. Three containers share one physical machine; the mounted repository replaces a distributed file system.

On the Windows host itself Spark 4 cannot read local Parquet files without the Hadoop native binaries, which the team chose not to install from unofficial sources; there data travel through Arrow, a path used only for tests and one benchmark row. Run times of the ETL are shown in Fig. 9: one to four cores speed the cluster up 2.6 times, but eight cores on two workers are no faster than four on one (40.5–41.2 s against 38.2 s).

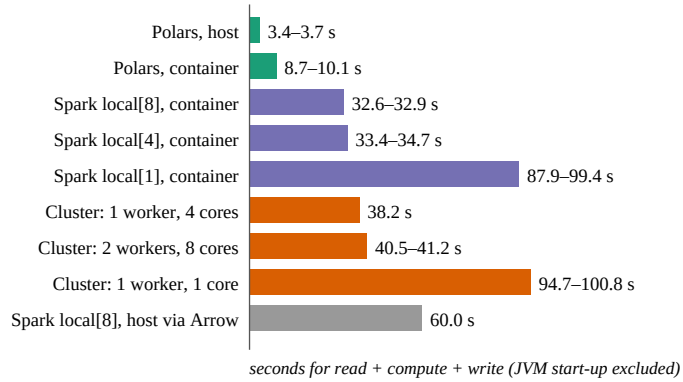


Fig. 9. Run time of the hourly-to-daily ETL (6,544,896 rows). Bars: mean of the runs; labels: minimum and maximum. Green: Polars; purple: Spark local mode; orange: standalone cluster; grey: Spark on the host via Arrow.

B. Full-scale satellite features

Report 4 also ran the hourly-feature job on the full satellite record (101.0 million hourly rows, 125 MB of Parquet; Table IX). On the two-worker cluster all 6,246,870 compared values (252,480 hourly rows and 54 daily features per product) are bit-identical to Polars. Spark is still slower: in medians, the best Spark set-up (local[8]) takes 1.8, 4.7 and 5.0 times as long as Polars in the same container, and the second worker does not

help, because both workers share one machine and its disk.

TABLE IX
FULL-RECORD HOURLY SATELLITE FEATURES IN THE CONTAINERS: SECONDS (MIN–MAX OVER RUNS; FIRST CLUSTER RUN WITH COLD CACHE), START-UP EXCLUDED; LOCAL[8] WITH A 4 GB DRIVER

Product	Hourly rows	MB	Cluster 2 x 4	Cluster 1 x 4	local[8]	Polars
IMERG	50,477,000	47.7	37.8–67.9	39.1–65.5	29.7–29.9	12.4–20.3
GSMaP-GC	25,241,800	50.8	30.5–62.3	29.8–51.1	23.5–24.7	4.8–5.4
GSMaP_NRT	25,245,600	26.3	29.4–71.0	29.2–31.4	23.2–23.3	4.7

C. MLlib models

MLlib gradient-boosted trees (GBT) and random forests (RF) were trained on the cluster with the configuration-A panel and the same temporal split; Spark's CrossValidator splits rows at random, which lets a model see the future, so it was replaced by a loop over a fixed validation period (36 fits). LightGBM stays best one day ahead (Table X); RF is ahead at h = 2–3 on one split without a significance test, and all three cut peaks by a similar amount.

TABLE X
MLLIB ON THE TWO-WORKER CLUSTER AGAINST LIGHTGBM, SAME PANEL AND SPLIT

Model	NSE h=1	h=2	h=3	Peak bias h=1	h=2	h=3
LightGBM, config. A	0.747	0.420	0.175	–0.45	–0.76	–0.89
Spark GBT	0.699	0.429	0.182	–0.50	–0.74	–0.87
Spark RF	0.687	0.454	0.214	–0.52	–0.75	–0.87

D. Deep learning

An LSTM [11] reads 14 or 30 days of nine raw daily variables and predicts the three horizons jointly; an MLP on the LightGBM lag columns is the deep baseline (8 configurations, three seeds). Training was distributed with PyTorch DDP [22] through Spark, both with TorchDistributor and with a barrier stage written for the project (TorchDistributor fails on Windows because the PyTorch build lacks libuv). The single-process run reproduces the experiment exactly; all two-rank runs give the same result (0.712, 0.413, 0.160) on one machine and on two workers; DDP changes NSE by +0.003, –0.019 and –0.020, about one seed standard deviation; and it is slower (9.9–10.1 s against 9.0–9.5 s for one process), because a model of 19,395 parameters spends more on gradient synchronisation than the split saves.

TABLE XI
DISTRIBUTED LSTM TRAINING (L = 30, MSE, SEED 42); RANGES OVER REPEATED RUNS

Run (where)	Proc. × threads	Global batch	Best epoch	train() (s)	Wall (s)
Single process (host)	1 × 4	64	10	16.1–18.2	16.1–18.2
Barrier, local[2] (host)	2 × 2	128	13	17.6–17.7	26.3–26.8
TorchDistributor, local (host)	2 × 2	—	—	failed: no libuv	—
Single process (container)	1 × 4	64	10	9.0–9.5	9.0–9.5
TorchDistributor, local (container)	2 × 2	128	13	9.9–10.1	16.3–16.4
TorchDistributor, 2 workers (cluster)	2 × 4	128	13	9.9–10.1	22.6–23.2
Barrier, 2 workers (cluster)	2 × 4	128	13	9.9–10.4	15.0–15.5

On the single split some deep configurations beat LightGBM at h = 2 and 3 (best: a peak-weighted MLP, 0.490 against 0.420 at h = 2), but the LSTM selected on validation lost at all horizons (0.709, 0.404, 0.171 against 0.747, 0.420, 0.175); the better configurations were identified only after seeing the test set.

Under the walk-forward (Table XII, Fig. 10) no deep model beats LightGBM consistently: the LSTM with MSE loss is level at h = 1 (+0.021, interval includes zero) and the peak-weighted LSTM loses clearly at h = 3 (–0.229, p = 0.033). Deep models detect more level-1 days (pooled CSI 0.49 against 0.44) and cut peaks less, but their seed-to-seed spread is 0.03–0.20 NSE against at most 0.01 for LightGBM.

TABLE XII
DEEP LEARNING AGAINST LIGHTGBM (PANEL A), 13 YEARLY FOLDS: ΔNSE MEAN, 95% INTERVAL, FOLDS WON/LOST

Model	h = 1	h = 2	h = 3	CSI lv. 1, h=1
LightGBM (NSE)	0.743	0.334	0.118	0.44
LSTM L=30, MSE	+0.021 [–0.014; +0.058] 8/5	–0.025 [–0.117; +0.032] 7/6	–0.026 [–0.101; +0.021] 7/6	0.49
LSTM L=30, peak-weighted	–0.007 [–0.065; +0.037] 8/5	–0.107 [–0.239; –0.008] 5/8	–0.229 [–0.411; –0.073] 5/8	0.49
MLP on lags, peak-weighted	+0.000 [–0.031; +0.037] 5/8	–0.065 [–0.142; +0.002] 5/8	–0.118 [–0.255; –0.015] 6/7	0.49

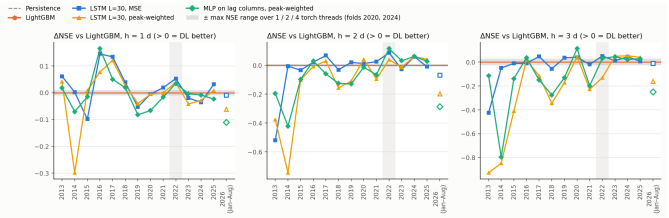


Fig. 10. Deep learning against LightGBM under the walk-forward: ΔNSE per test year, positive when the deep model is better; grey band: largest change caused by the torch thread count.

E. Task parallelism and honest assessment

The deep-learning walk-forward needed 168 independent fits (4,049 s in sequence); a local pool of 6 processes finished them in about 13 minutes. This task parallelism, not data-parallel DDP, is what shortened the experiments. For the peak experiments of Section VII, 141 LightGBM runs took 329.9 s on a pool of 8 processes. The same runs as Spark tasks (`spark_walkforward.py`, two workers, 8 cores) took 270.7 s, 7.7 times faster than the summed task time, and reproduced the local forecasts of 141 runs up to rounding (largest difference 2.3×10^{-13} m^3/s; LightGBM is built differently for Linux and Windows). Part of the gap to the Windows pool may come from the operating system and the LightGBM build rather than from Spark; what is established is near-linear scaling of independent fits on the cluster.

At this data size Spark is slower than Polars: the two-worker ETL takes 11.6 times as long as Polars on the host, and the hourly features 7.3 times on the cluster. Measured causes are reading through the Windows–Docker mount, partitions skewed by only 64 cell keys, fixed start-up costs and two workers sharing one machine. Spark becomes the right tool when data no longer fit one machine or already live in distributed storage, for example many river cells across Vietnam, or folds and seeds spread over several real machines [23]. The code is portable: the session reads the master address from an environment variable and the training function takes the panel path as a parameter, so a managed workspace such as Databricks [31] could host the same jobs with Delta tables. That step is a proposal; it was not run and no result here comes from it.

VII. EXPERIMENTS ON PEAK UNDERPREDICTION

Report 4 also tested three remedies for peak cutting on the 13-fold walk-forward with criteria fixed before running: accept a variant only if ΔNSE at h = 2 or 3 is positive with p below 0.05 and h = 1 loses at most 0.01, or if detection rises clearly at that cost. (B1) peak-weighted training, $w = 1 + \alpha \cdot \max(0, y/\text{level } 1 - 1)$; (B2) isotonic correction of log forecasts fitted on earlier folds, on all forecasts or only on high ones; (B3) a fixed 50/50 average of E and the LSTM. The baseline reproduces Report 3 (0.761, 0.437, 0.206).

TABLE XIII
PEAK EXPERIMENTS, 13 YEARLY FOLDS: ΔNSE AGAINST E (FOLDS WON/LOST, WILCOXON p)

Variant	ΔNSE h = 1	h = 2	h = 3	POD h=1
E (baseline NSE)	0.761	0.437	0.206	0.55
B1, α = 1	+0.007 (8/5, p 0.59)	–0.001 (5/8, p 0.89)	–0.003 (6/7, p 0.54)	0.59
B1, α = 3	+0.011 (9/4, p 0.13)	+0.012 (9/4, p 0.22)	+0.001 (6/7, p 0.89)	0.62
B1, α = 5	+0.008 (6/7, p 0.54)	+0.005 (7/6, p 0.74)	–0.007 (5/8, p 0.34)	0.62
B1, α = 10	–0.010 (4/9, p 0.45)	+0.001 (7/6, p 1.00)	–0.008 (4/9, p 0.34)	0.53
B2, isotonic	–0.040 (6/7, p 0.64)	–0.008 (5/8, p 0.50)	–0.027 (1/12, p 0.001)	0.57
B2, high only	–0.042 (6/7, p 0.59)	–0.005 (7/6, p 0.74)	–0.013 (4/7, p 0.067)	0.57
B3, E + LSTM	**+0.029 (10/3, p 0.048)**	**–0.022 (2/11, p 0.068)**	**–0.023 (2/11, p 0.021)**	**0.57**

No variant meets the criteria, so E remains the main model. Peak weighting with α = 3 lifts h = 1 slightly (+0.011 [+0.000; +0.024], p = 0.13), POD from 0.55 to 0.62 and the h = 1 peak bias from –0.27 to –0.21, but the gain is not significant, not monotone in α and α was widened after seeing the first results; the h = 2–3 peak bias (–0.65, –0.82) does not move. Isotonic correction hurts (h = 3: –0.027 [–0.040; –0.016], 1/12 folds, p = 0.001) because the mapping from forecast to target is unstable across years. The E + LSTM average gives the only significant improvement of the whole series, at h = 1 (+0.029 [+0.006; +0.056], 10/3 folds, p = 0.048), but it is worse at h = 2 (–0.022) and h = 3 (–0.023, p = 0.021). Using it only at h = 1 would be a choice made after seeing the results and would add PyTorch to the daily job; it is recorded as future work, to be tested on new data. Two to three days before a peak the flood-producing rain has usually not fallen, so peak cutting at h = 2–3 needs good rainfall forecasts or release data, not a different loss.

VIII. OPERATIONAL SYSTEM

A GitHub Actions job runs every morning (scheduled at 20:00 UTC, in practice near 05:00 local time). It downloads the issued GloFAS forecast and weather forecast, applies the reservoir floor rule, writes one CSV per issue date to `reports/forecast_log/` and commits it; a check fails the job if a day is missing. The log starts on 15 September 2026 and holds 48 issue dates up to 2026–11–02 (20 September was lost to a time-zone bug and is not imputed).

The job now also runs the frozen main model: 88 predictors, three seeds per horizon, L1 at h = 1 and L2 at h = 2–3, refit on all labelled days from 1998 to August 2026 (10,466 rows at h = 1). It writes `lgbm_q`, the floored `lgbm_san_ho` and a risk level, plus `p_level1`: the probability that tomorrow's discharge reaches 806 m^3/s, calibrated by isotonic regression on 9,862 out-of-fold days of 1999–2025 (185 positive), with an alert at 0.091 = 1/11. Models are plain-text LightGBM files

downloaded from a GitHub release; the calibration is applied with linear interpolation, so the job needs neither PyTorch nor scikit-learn.

Live inputs differ from training inputs. The training panel uses ERA5 reanalysis; the job uses ERA5 up to about six days before the issue date and Open-Meteo forecast data after that, so rain on day t is a *forecast* for the whole day. The noise experiment of Report 3 ($\sigma = 1$ lowers NSE at h = 1 from 0.742 to 0.683) bounds the expected loss. Soil moisture is held at its last ERA5-Land value, a failed IBTrACS download is treated as no storm and flagged, and the GloFAS value for day t is the latest forecast, not the analysis.

The first operational run with the LightGBM columns was issued on 2026–11–02 (1 run(s) so far); AC-3 still needs weeks of forecasts overlapping verifying GloFAS values.

Dashboard. A Streamlit application reads the log and has three tabs: the forecast (30 past days and three forecast days for raw GloFAS, floored GloFAS and LightGBM, with the three risk-level lines and a table of levels), a map of the five sub-basins coloured by the basin-wide level at Kim Long, and an explanation tab with grouped SHAP shares, skill scores and limitations. It states on every page that the product is academic and that risk levels are not the official alert stages.

IX. DISCUSSION

What the correction layer adds. The model does not see a better river than GloFAS; it sees the same simulated river earlier. Its gain over persistence comes from information GloFAS does not use at the issue time: the antecedent state carried by recent discharge, rain already fallen on the basin, and synoptic drivers (storms, cold-surge pressure gradients) that announce rain one to three days ahead. That is why the large-scale predictors helped at two and three days while soil moisture and storage proxies did not, and why skill collapses when the flood-producing rain has not yet fallen.

Single split against walk-forward. Several conclusions changed when the single test period was replaced by thirteen yearly folds. IMERG looked useful at h = 2 on the single split (+0.017) but its fold gain (+0.013) has an interval that includes zero; deep models that beat LightGBM on the test period did so only in the years that period happened to contain; and the loss switch for h = 2–3 was decided on folds and validation, never on the test set. Every acceptance decision in this project therefore rests on the walk-forward, and the criteria for the peak experiments were written down before running them.

Negative results as results. Five lines of work did not improve the operational forecast: reservoir records, satellite rainfall, deep learning, peak weighting and Spark. Each produced something usable nonetheless: the reservoir tests justify percentile risk levels and the floor rule; the latency test showed which satellite product could be used live; the deep models are the best detectors at one day; and the Spark jobs reproduce the pipeline exactly, ready for larger inputs. Table XIV lists where the project departed from the proposal and why.

TABLE XIV
DEVIATIONS FROM THE PROPOSAL

Proposal (Report 1)	Delivered	Reason
Official alert stages as labels	Percentile risk levels	Discharge does not determine the water level
Raw GloFAS forecast as baseline	Forward log since Sept. 2026	Past flood forecasts are not archived
Scenario B with archived rain forecasts	Noise test on rainfall	The archive returned ERA5 values
Three latitude bands	Five DEM sub-basins	Hydrologically meaningful units
Commune-level risk map	Sub-basin map, one basin-wide level	Target exists at one cell only
Fixed 0.5 alert threshold	Cost-based, calibrated, 1/11	Misses cost more than false alarms
LSTM only if time allowed	LSTM, MLP, distributed DDP	Requested in the review of Report 3
Single machine	Spark cluster, MLlib, task parallelism	Requested in the review of Report 3

X. LIMITATIONS AND FUTURE WORK

- **Simulated target.** Scores measure agreement with GloFAS, which itself under-simulates the largest floods; true peaks are under-estimated twice. GloFAS v5 adds large reservoirs and may change the target; the dam-era test must then be repeated.
- **h = 3 below target.** AC-1 is missed at three days (0.236 test, 0.206 walk-forward against 0.25). Archived rainfall forecasts would be the most promising input; the archive tested in Report 2 returned ERA5 values and could not be used.
- **Regime change.** The test period consists of archived GloFAS forecasts; only 3 walk-forward years fall in that regime, for which the smallest attainable Wilcoxon p is 0.25, so it cannot be tested alone. In 2023–2025 configuration E scored 0.772, 0.522 and 0.223, in line with earlier years.
- **One cell, daily target.** The 5 km grid does not resolve the Bo River, so only Kim Long is forecast; the sub-basin map shows one basin-wide level. Finer and hourly inputs cannot make the daily, gridded target finer.
- **Resolution of comparisons.** NSE differences below about 0.01 are numerical noise; conclusions rest on fold-level intervals, not on means.
- **Rare events.** Only level 1 has enough events; level 2 and AC-2 cannot be evaluated, and risk levels are relative, not official stages.
- **No raw-GloFAS comparison yet.** AC-3 depends on the forecast log, which is short and has not yet overlapped with verifying values.
- **Operational inputs.** Live rainfall for day t is a forecast; satellite rainfall gave no operational gain on configuration E.
- **Future work:** a horizon-specific E + LSTM ensemble at h = 1 tested on new seasons; walk-forward folds as Spark

tasks on several machines; requesting gauge data if the project continues beyond the course.

XI. CONCLUSION

Open global data support a useful local forecast of GloFAS discharge at Kim Long. LightGBM with 88 predictors reaches NSE 0.747, 0.499 and 0.236 on the test period and 0.761, 0.437 and 0.206 across 13 years, far above persistence beyond one day, catches level-1 flood events one day ahead as often as persistence but with lead time, and provides calibrated alert probabilities. It cuts flood peaks and misses the three-day target, for reasons that were measured rather than tuned away. Large-scale weather predictors helped; reservoir data, satellite rain, deep learning, peak weighting and Spark did not improve the operational forecast at this scale, and each negative result is documented with its evidence. The model now runs daily, and its forecasts are shown on a dashboard that states its limits. The research questions of Report 1 are answered as follows.

- **RQ1.** Yes at one and two days (NSE 0.747 and 0.499, RMSE 25.6% below persistence at one day); only partly at three days (0.236), with peaks cut at all horizons.
- **RQ2.** Not against official stages, which are water levels that simulated discharge does not determine; against the first risk level the classifier reaches POD 0.645 and FAR 0.200 one day ahead, and the calibrated cost-based rule POD 0.88 with FAR 0.54 across folds.
- **RQ3.** Only indirectly: the target exists at Kim Long alone, but SHAP gives upper-basin rainfall the largest rain share one day ahead (11.2% against 9.5% for the basin mean), consistent with the two reservoir catchments supplying most of the flow.

TABLE XV
TEAM ROLES

Member	Role	Main contributions
Dang Nguyen Giap	Tech Lead	Data engineering, modelling and evaluation, Spark and deep learning, daily job, integration and review
Ho Anh Duc	Data Analyst	Exploratory analysis, feature engineering, baselines, dashboard
Mai Thi Le Huyen	Research & Docs	Literature review, legal sources, data dictionary, report editing and slides

REFERENCES

[1] L. Alfieri *et al.*, “GloFAS – global ensemble streamflow forecasting and flood early warning,” *Hydrol. Earth Syst. Sci.*, vol. 17, no. 3, pp. 1161–1175, 2013.

[2] S. Harrigan *et al.*, “Daily ensemble river discharge reforecasts and real-time forecasts from the operational Global Flood Awareness System,” *Hydrol. Earth Syst. Sci.*, vol. 27, no. 1, pp. 1–19, 2023.

[3] H. Hersbach *et al.*, “The ERA5 global reanalysis,” *Q. J. R. Meteorol. Soc.*, vol. 146, no. 730, pp. 1999–2049, 2020.

[4] Prime Minister of Vietnam, “Decision 05/2020/QD-TTg on water levels corresponding to flood alert levels,” *Official Gazette*, no. 179+180, Feb. 2020, Annex I, row 135. [in Vietnamese]

[5] H. S. Nguyen *et al.*, “Synoptic patterns causing heavy rain and floods in Thua Thien Hue province in 2020,” *Hue Univ. J. Sci.: Earth Sci. Environ.*, vol. 131, no. 4A, pp. 149–162, 2022. [in Vietnamese]

[6] G. Nearing *et al.*, “Global prediction of extreme floods in ungauged watersheds,” *Nature*, vol. 627, pp. 559–563, 2024.

[7] V. Nourani, S. Kheirieh, S. A. Kantoush and J. J. Huang, “Bias correction of Global Flood Awareness System (GloFAS) data for multi-station river flow prediction by ensemble modelling,” *Eng. Appl. Comput. Fluid Mech.*, vol. 20, no. 1, 2665857, 2026.

[8] T. Honcharenko *et al.*, “Explainable deep ensemble bias correction of GloFAS-ERA5 streamflow across snow-influenced transboundary basins of Central Asia,” *Water*, vol. 18, no. 16, 2055, 2026.

[9] S. Gnjato, I. Lescesen, X. Zhou and D. Dukanovic, “Explainable machine learning for streamflow forecasting: application to the Bosna River basin,” *Water*, vol. 18, no. 10, 1226, 2026.

[10] G. Ke *et al.*, “LightGBM: A highly efficient gradient boosting decision tree,” in *Adv. Neural Inf. Process. Syst.*, vol. 30, 2017, pp. 3146–3154.

[11] F. Kratzert, D. Klotz, C. Brenner, K. Schulz and M. Herrnegger, “Rainfall–runoff modelling using Long Short-Term Memory (LSTM) networks,” *Hydrol. Earth Syst. Sci.*, vol. 22, no. 11, pp. 6005–6022, 2018.

[12] F. Kratzert, D. Klotz, G. Shalev, G. Klambauer, S. Hochreiter and G. Nearing, “Towards learning universal, regional, and local hydrological behaviors via machine learning applied to large-sample datasets,” *Hydrol. Earth Syst. Sci.*, vol. 23, no. 12, pp. 5089–5110, 2019.

[13] J. H. Ougahi and J. S. Rowan, “Investigating deep learning knowledge transfer in streamflow prediction from global to local catchment,” *Water Resour. Res.*, 2026, doi: 10.1029/2025WR041194.

[14] J. E. Nash and J. V. Sutcliffe, “River flow forecasting through conceptual models part I – a discussion of principles,” *J. Hydrol.*, vol. 10, no. 3, pp. 282–290, 1970.

[15] H. V. Gupta, H. Kling, K. K. Yilmaz and G. F. Martinez, “Decomposition of the mean squared error and NSE performance criteria: implications for improving hydrological modelling,” *J. Hydrol.*, vol. 377, no. 1–2, pp. 80–91, 2009.

[16] D. N. Moriasi *et al.*, “Model evaluation guidelines for systematic quantification of accuracy in watershed simulations,” *Trans. ASABE*, vol. 50, no. 3, pp. 885–900, 2007.

[17] G. W. Brier, “Verification of forecasts expressed in terms of probability,” *Mon. Weather Rev.*, vol. 78, no. 1, pp. 1–3, 1950.

[18] B. Zadrozny and C. Elkan, “Transforming classifier scores into accurate multiclass probability estimates,” in *Proc. 8th ACM SIGKDD Int. Conf. Knowl. Discov. Data Min.*, 2002, pp. 694–699.

[19] S. M. Lundberg and S.-I. Lee, “A unified approach to interpreting model predictions,” in *Adv. Neural Inf. Process. Syst.*, 2017, pp. 4765–4774.

[20] M. Zaharia *et al.*, “Apache Spark: A unified engine for big data processing,” *Commun. ACM*, vol. 59, no. 11, pp. 56–65, 2016.

[21] X. Meng *et al.*, “MLlib: Machine learning in Apache Spark,” *J. Mach. Learn. Res.*, vol. 17, no. 34, pp. 1–7, 2016.

[22] S. Li *et al.*, “PyTorch distributed: Experiences on accelerating data parallel training,” *Proc. VLDB Endow.*, vol. 13, no. 12, pp. 3005–3018, 2020.

[23] F. McSherry, M. Isard and D. G. Murray, “Scalability! But at what COST?,” in *Proc. 15th Workshop Hot Topics Oper. Syst. (HotOS XV)*, 2015.

[24] ESA/Copernicus, “Copernicus DEM GLO-30.” [Online]. Available: https://copernicus-dem-30m.s3.amazonaws.com/

[25] Prime Minister of Vietnam, “Decision 1606/QD-TTg on the inter-reservoir operating procedure for the Huong River basin,” Nov. 2019. [in Vietnamese]

[26] Open-Meteo, “Flood API and Historical Weather API,” 2026. [Online]. Available: https://open-meteo.com/en/docs

[27] NOAA NCEI, “International Best Track Archive for Climate Stewardship (IBTrACS) v04r01.” [Online]. Available: https://www.ncei.noaa.gov/products/international-best-track-archive

[28] NOAA CPC, “Oceanic Niño Index.” [Online]. Available: https://www.cpc.ncep.noaa.gov/data/indices/oni.ascii.txt

[29] NASA, “GPM IMERG V07, Earth Engine collection NASA/GPM_L3/IMERG_V07.”

[30] JAXA, “GSMaP v8 operational, Earth Engine collection JAXA/GPM_L3/GSMaP/v8/operational; GSMaP_NRT, JAXA G-Portal.”

[31] Databricks, “Databricks Free Edition,” documentation. [Online]. Available: https://docs.databricks.com/aws/en/getting-started/free-edition